MAHFUZUR RAHMAN EMON

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PROJECT EXPERIENCE

EDUSTREAM+ - Cloud-native learning platform (Node.js, Express, Azure, Blob Storage, Table Storage) | GitHub
Independent full-stack project | Azure App Service, Key Vault, Application Insights, GitHub Actions CI/CD

- Full-stack platform where educators publish courses that combine video lectures, illustrated notes and audio in a single course.
- Direct-to-blob uploads with scoped, time-limited SAS tokens handle 1GB+ videos without request timeouts.
- Two-tier content moderation (Azure AI Content Safety with a local fallback) screens all user-submitted text.
- Secrets held in Azure Key Vault via Managed Identity, with Helmet, rate limiting and Application Insights telemetry; deployed through GitHub Actions.

SMART EYE - Ocular health screening platform (Python, FastAPI, TensorFlow/Keras, React) | GitHub
University capstone | Deep-learning CNNs, fuzzy risk fusion and Grad-CAM explainability

- Fuses three AI pipelines (fundus and anterior-segment CNN classifiers plus real-time fatigue detection) into one Ocular Health Index.
- Grad-CAM heatmaps show which image regions drove each prediction, keeping results explainable rather than black-box.
- A Mamdani fuzzy inference system, implemented in pure NumPy, fuses confidence, fatigue and symptom intensity into the final risk score.
- React and Vite front end over a FastAPI REST API, with SQLite sessions, PDF report export and a bilingual English/Bengali interface.

CAREFLOW AI - NHS appointment prioritisation copilot (React, TypeScript, Vite, TanStack, Tailwind) | GitHub
Independent product project | Separate doctor and patient views, deployed on Vercel

- Decision-support copilot that prioritises NHS appointments and referrals across wards and patient queues.
- Separate doctor and patient interfaces built as a typed React single-page app with TanStack Router and Query.
- Form flows validated end-to-end with react-hook-form and Zod; metrics visualised with Recharts.
- Accessible component system built on Radix UI and Tailwind CSS, shipped to Vercel.

SMART STUDY PLANNER - Full-stack study workspace (Flask, MongoDB, Angular 17, Chart.js) | GitHub
Independent full-stack project | REST API, JWT auth and analytics dashboards

- Organises learning as Subject to Topic to Subtopic with exercises, notes and resources, plus full-text search across all of them.
- Flask REST API backed by MongoDB, with JWT sessions, bcrypt password hashing and optional Auth0 social login.
- Angular 17 front end using standalone components, signals and Reactive Forms, with a live progress dashboard.
- Three Chart.js analytics views and an admin dashboard scoped to admin users only.

TECHNICAL SKILLS

Full-stack development: Python, TypeScript, JavaScript, React, Angular, Node.js, Express, FastAPI, Flask, Vite and REST APIs.
Cloud and data: Microsoft Azure (App Service, Blob Storage, Table Storage, Key Vault, Application Insights), MongoDB, SQLite, Managed Identity and SAS tokens.
AI and machine learning: TensorFlow and Keras, CNNs (ResNet-50, VGG-16, EfficientNet), scikit-learn, NumPy, Grad-CAM, fuzzy inference, RAG pipelines and vector stores.
Quality and delivery: Git, GitHub Actions, CI/CD, pytest, JWT and OAuth (Auth0, Google), Helmet, rate limiting, content moderation and technical documentation.

PROFESSIONAL EXPERIENCE

Open-Source Contributor | OWASP OpenShield (Azure CSPM) | 2026

- Contributor to OpenShield, an OWASP-listed open-source Cloud Security Posture Management (CSPM) tool for Azure, through pull requests merged after core-team review.
- Added an Azure Privileged Identity Management detection rule (AZ-IDN-004) flagging admin roles without PIM, together with its remediation playbook.
- Built an AI document-ingestion and vector-store RAG pipeline (AI-004) supporting the platform's AI features.
- Made the Flask API test-safe and wired the full pytest suite into the GitHub Actions CI pipeline.

EDUCATION AND ADDITIONAL INFORMATION

Ulster University, London, UK
BSc (Hons) Computing Systems | Awarded September 2026 | First Class Honours
Awards & recognition: Dean’s List Year 1 & 2
Certifications: Python for Beginners by QA LTD, Career Essentials in Generative AI by Microsoft and LinkedIn
Availability: Full-time, in London, available from September 2026.
Leadership & contribution: Student Representative and SSCC Member (January 2024 - September 2026): Represented students, raised their concerns, and took part in discussions about modules and the course.